

1 Question (20 points) (Java language)

Suppose arrays representing sets A and B are both sorted. Write a linear time method that finds if A is a subset of B.

1 `boolean isSubset(int[] A, int[] B)`

2 Question (15 points) (C++ language)

Suppose you are given an array of N integers sorted in increasing order. Write a linear time method to convert this array sorted in decreasing order.

1 `void convertDecreasingOrder(int* A, int N)`

3 Question (15 points) (Java language)

```
1 class Node {  
2     int data;  
3     Node next;  
4 }  
5 class LinkedList {  
6     Node head;  
7     Node tail;  
8 }
```

Write a linear time class method in LinkedList class to delete the nodes indexed between p and q (including p'th and q'th items) from a singly linked list.

```
1 void deleteBetween(int p, int q)
```

4 Question (15 points) (C++ Language)

```
1 class Tree {  
2     TreeNode* root;  
3 }  
4 class TreeNode {  
5     TreeNode* left;  
6     TreeNode* right;  
7     int data;  
8 }
```

Write a class method in `TreeNode` class that computes the products of all keys in a binary search tree.

```
1 int productOfTree()
```

5 Question (20 points) (Java language)

```
1 public class Edge {  
2     int from;  
3     int to;  
4     int weight;  
5     Edge next;  
6 }  
7 public class EdgeList {  
8     Edge head;  
9     Edge tail;  
10 }  
11 public class Graph {  
12     int vertexCount;  
13     EdgeList[] edges;  
14 }
```

A bipartite graph is a graph such that vertices of the graph can be partitioned into two subsets such that no edge has both its vertices in the same subset. Write a class method in Graph class for adjacency list representation which checks if the corresponding graph is bipartite or not. Hint: Use Depth or breath first search to traverse the graph.

```
1 boolean isBipartite ()
```

6 Question (15 points)

Given the following function

```
1 int magic(int N){  
2   if (N == 0)  
3     return 0;  
4   int sum = 0;  
5   for (i = 0; i < N; i++)  
6     sum++;  
7   return magic(N / 2) + sum;  
8 }
```

What is the time complexity of magic? Extract the recurrence equation for the time complexity. Solve the recurrence equation without the master theorem.